

Mohammed Ali

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EDUCATION

University of Florida

B.S. Computer Science

GPA: 3.62/4.0 **Dean's List:** Spring 2024, Fall 2023, Spring 2023

Relevant Coursework: Data Structures and Algorithms, Discrete Structures, Linear Algebra

Gainesville, FL

May 2026

TECHNICAL SKILLS

Game Engines: Unreal Engine 5 (Blueprint & Gameplay Systems), Unity, Godot

Programming: C++, Python, JavaScript / TypeScript, GDScript

Tools: Git, GitHub, Blender, Niagara, Express.js, Node.js, React Native, Jest

Concepts: Lock-Free Data Structures, Concurrent Programming, Gameplay Systems, Event-Driven Design

SYSTEMS PROJECTS

Lock-Free Concurrent Queue — C++17

Solo Project

- Implemented the Michael-Scott non-blocking queue from scratch using `std::atomic` and CAS loops, guaranteeing lock-free progress across concurrent producer and consumer threads.
- Eliminated false sharing by placing `head_` and `tail_` on separate 64-byte cache lines; resolved the ABA memory hazard using a bump-allocated node pool that avoids unsafe `delete` under concurrent access.
- Benchmarked against a mutex baseline across three scenarios with up to 8 threads and 1.6M operations; documented results and analysis in a GitHub README.
- Wrote concurrent correctness tests covering FIFO ordering, empty-queue semantics, 8-thread producer races, and interleaved producer-consumer workloads.

GAME DEVELOPMENT PROJECTS

Re:Sigil — Psychological Horror Game (Unreal Engine 5)

Solo Project

- Designed and implemented a UE5 vertical slice featuring a modular interaction system, event-driven world progression, puzzle mechanics, and narrative UI systems.
- Implemented event-driven gameplay logic controlling puzzles, door states, and story progression.
- Created a world-space subtitle UI system with typewriter effects for immersive dialogue.
- Integrated Niagara particle systems and dynamic lighting to support atmospheric horror gameplay.

Draco Disease — RPG Game (Unreal Engine 5)

Game Jam

- Developed gameplay mechanics for a team-based RPG during a game jam.
- Implemented a cursed altar interaction granting buffs with randomized debuffs.
- Integrated toon shading outlines to maintain stylistic consistency between 2D and 3D elements.

AI RPG Game — Godot + Python

- Built a prototype RPG integrating AI-generated abilities using the OpenAI API.
- Designed an algorithm dynamically assigning combat statistics to generated abilities.

EXPERIENCE

University of Florida

Discrete Mathematics Teaching Assistant

- Led discussion sessions reinforcing lecture material and held weekly office hours for undergraduate students.

Kyoto University

Architecture Research Assistant

- Modeled complex Monge surface geometries using Rhino and Grasshopper; contributed to a research paper exploring mathematical surface applications.

LANGUAGES

English (Native) | Arabic (Native) | Japanese (Advanced)